FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.

PHASE 1 SCOPE OF WORK & TECHNICAL-COMMERCIAL PROPOSAL

Comparative Diagnostic R&D Study of Tanzanian Imported Split Toor Dal vs. Indian Benchmark Toor Dal

Project Reference & Document Ref:	FNQ-PR-2026-001 | SOW Rev V10 (Evidence-Gated Baseline)
Client Entity & Principal:	Technical Advisory Panel (Kolkata – 700006, West Bengal, India)
Client Contact & Designation:	Technical Advisory Panel, CEO & Director
Executing R&D Contractor:	FinnoaQ Agro-Industrial Technologies Ltd.
Lead Chemical & Process Engineer:	Sourish Senapati (Lead Process Engineer, FinnoaQ)
Research Advisory & Oversight:	Technical Advisory Panel & FinnoaQ Process Engineering Board
Issue Date & Commercial Status:	20 October 2026 | Rectified Technical Proposal Issued for Client Commissioning

1. Executive Decision Summary

This technical-commercial proposal is submitted by FinnoaQ Agro-Industrial Technologies Ltd. ('FinnoaQ') to the Client to initiate a structured, evidence-gated Phase 1 Comparative Diagnostic R&D Study. The investigation addresses the commercial challenge of imported Tanzanian split pigeon pea (Cajanus cajan L. Millsp., 'toor dal'), which reportedly exhibits substantial cooking latency compared to domestic Indian benchmark dal under household preparation conditions.

Crucial Diagnostic Foundation (Client Feedback A): In prior technical drafts, pectin de-esterification and divalent cation crosslinking were emphasized as modeled mechanisms. In this rectified Rev V10 baseline, pectin-mineral interactions are rigorously repositioned as one hypothesis (Hypothesis H3) among ten competing physical, physiological, genetic, and processing hypotheses. The primary objective of Phase 1 is not to assume an answer or commercialize an unverified chemical recipe, but to conduct an objective diagnostic investigation designed to systematically discriminate between competing hypotheses on the client's actual grain lot.

Independent Mass Balance Audit (Client Feedback B): The downstream 15,000 kg/day processing concept has been independently audited from first principles. For an incoming feed of 15,000.000 kg/day at 10.5% wet-basis moisture (13,425.000 kg/day dry solids), solids losses are rigorously reconciled between the audited 184.000 kg/day (1.3706%) baseline and the alternate 196.005 kg/day (1.46%) assumption. Furthermore, the pre-dryer moisture discrepancy—between 28.000% wet-basis (3,678.056 kg/day evaporation) and 28.7735% wet-basis implied by the 18,590 kg/day stream (3,877.778 kg/day evaporation)—is quantified as a delta of 199.722 kg/day water (+5.43% thermal drying duty) and explicitly designated as UNRESOLVED EMPIRICALLY pending physical screen drainage measurements.

Core Research Problem	Tanzanian imported toor dal exhibits prolonged cooking delay (~6-8 household whistles vs ~3 whistles for Indian benchmark).
Phase 1 Scientific Objective	Standardized diagnostic comparison across 9 analytical modules to isolate rate-limiting mechanisms and discriminate 10 hypotheses.
Proposed Professional Fee	₹1,73,000 plus applicable GST (fixed-price diagnostic scope; commercial ceiling up to ₹2,00,000 + GST).
Indicative Turnaround	20–25 business days across 5 structured operational stages upon sample receipt and project mobilization.

Key Deliverables	14 discrete deliverables including 9 analytical reports, benchmark matrix, root-cause assessment, and Phase 2 roadmap.
Action Requested from Client	Approve Phase 1 Scope of Work (Option 1 baseline or Option 2 multi-lot), authorize Milestone 1 advance (₹69,200 + GST), and dispatch sample lots.

[MANDATORY SCOPE BOUNDARY & SCIENTIFIC PRINCIPLE]
Phase 1 is strictly a diagnostic R&D investigation. It does not promise, warrant, or guarantee an immediate commercial product, an optimized chemical formulation, or a construction-ready industrial facility. Phase 1 delivers the empirical ground truth required to make defensible engineering decisions. All formulation development (Phase 2), pilot production (Phase 3), and commercial plant FEED (Phase 4) are separately scoped, budgeted, and authorized future milestones.

2. Problem Background & Commercial Context

Imported pulses represent a vital commodity buffer for the Indian milling, processing, and distribution ecosystem. Tanzanian split pigeon pea offers significant procurement price and freight advantages. However, commercial millers and wholesale traders frequently encounter severe cooking latency in Tanzanian dal consignments compared to domestic Indian varieties (e.g. Maharashtra Latur Desi or Karnataka Gulbarga). In household pressure-cooking observations, Tanzanian dal is reported to require approximately 6 to 8 whistles to reach tenderness, whereas Indian benchmark dal reaches an acceptable table texture in approximately 3 whistles.

Scientific Distinction on Whistle Counting: In domestic cooking, a 'whistle' is an acoustic pressure-relief event governed by cooker geometry, burner heat flux, venturi nozzle diameter, and deadweight mass. It is a qualitative, highly non-linear indicator that conflates vessel thermodynamics with grain softening kinetics. Phase 1 replaces qualitative whistling with instrumented time-temperature-pressure logging and objective texture analyzer load-cell shear/compression measurements (< 24.0 N tenderness threshold).

3. Diagnostic Objectives & Success Definition

The Phase 1 R&D study is designed to achieve four core diagnostic objectives:

1. • Standardized Comparative Profiling: Quantify physical geometry, proximate composition, starch pasting/gelatinization, hydration kinetics, microstructural integrity, pectin solubility, and mineral interactions for Tanzanian dal versus Indian benchmark.

2. • Objective Cooking Protocol: Establish a reproducible, instrumented cooking protocol utilizing a calibrated compression/shear endpoint to measure exact softening kinetics and hard-core persistence rates.

3. • 10-Hypothesis Discrimination: Systematically test ten competing scientific hypotheses to isolate primary rate-limiting mechanisms from secondary statistical associations.

4. • Commercial Gatekeeping: Deliver an auditable technical foundation and prioritized roadmap for Phase 2 treatment trials, protecting the client against premature capital investment in unproven processing technologies.

Scientific Definition of Success: Under Section 33 of the source SOW, success in Phase 1 is defined by the quality, reproducibility, and defensibility of the diagnostic evidence. Phase 1 is successful when it delivers definitive comparative data, quantified measurement uncertainty, and prioritized root causes—regardless of whether the Tanzanian dal demonstrates fast cooking in its unconditioned state.

4. Phase Architecture & Scope Boundaries

To protect project governance and ensure commercial clarity, the overall technology roadmap is structured into four distinct, sequentially gated phases. Only Phase 1 is offered for commissioning under this proposal.

Phase	Designation & Scope	Core Technical Deliverables	Commercial Status
Phase 1	Comparative Diagnostic R&D Study	Standardized baseline comparison, 9 analytical modules, objective cooking protocol, 10-hypothesis discrimination, Phase 2 roadmap.	CURRENT OFFERING (₹1,73,000 + GST; 20–25 Days)

Phase	Designation & Scope	Core Technical Deliverables	Commercial Status
Phase 2	Treatment Development & Mechanistic Validation	Bench-scale screening of conditioning formulations (bicarbonate, citrate, hydrothermal), dose-response, pH, washing, drying, storage stability.	Gated Future Scope (Budget defined post-Phase 1)
Phase 3	Pilot Scale-Up & Shelf-Life Verification	Continuous pilot trials, empirical effluent load, microbial hold-time validation, packaging qualification, commercial shelf-life testing.	Gated Future Milestone (Contingent on Phase 2)
Phase 4	15,000 kg/day Plant Design & Execution	Detailed FEED package, frozen PFD/P&IDs, equipment specifications, civil/structural layouts, procurement support, plant commissioning.	Long-Term Investment (Provisional conceptual annexure)

Explicit Exclusions from Phase 1 Scope: Unless approved via written change order, Phase 1 strictly excludes: (1) Formulating, optimizing, or validating commercial chemical treatments; (2) Conducting pilot-scale processing or machinery fabrication; (3) Guaranteeing finished cooking-time reduction; (4) Packaging or multi-month shelf-life stability trials; (5) Issuing construction-ready drawings or detailed FEED for the 15 MT/day facility; (6) Filing regulatory or FSSAI product clearance applications.

5. Overall Technical Approach & Work Packages

The Phase 1 investigation is executed through five inter-dependent Work Packages (WPs) structured to move logically from raw material verification to advanced root-cause synthesis:

Work Package	Title & Focus	Key Operational Activities & Methods
WP-01	Sample Intake, Physical & Proximate Screening	Visual inspection, foreign matter, 1000-grain weight, bulk density, seed-coat ratio, AOAC moisture, protein, fat, ash, and fibre.
WP-02	Starch & Thermal Gelatinization Profiling	Total starch, amylose/amylopectin ratio via Megazyme assay, Differential Scanning Calorimetry (DSC) for onset, peak, conclusion temp and enthalpy.
WP-03	Hydration Kinetics & Absorption Modeling	Gravimetric time-course water absorption at 25°C, 40°C, and 60°C (0–360 min), initial absorption rate, saturation capacity, and leaching losses.
WP-04	Structural, Microstructural & Pectin Analysis	Cotyledon cellular density, seed-coat thickness, chemical fractionation of water-, oxalate-, and alkali-soluble pectin, degree of esterification, ICP-OES mineral profiling.
WP-05	Standardized Cooking, Texture & Root-Cause Synthesis	Instrumented pressurized cooking trials, objective texture analyzer compression/shear force, hard-core rate scoring, multivariate correlation, Phase 2 roadmap.

5.3 In Silico Digital Simulation Methodology & Decision-Support Framework

To maximize analytical rigor and protect client capital prior to laboratory expenditure, FinnoaQ deploys an offline, GPU-accelerated computational engineering methodology as a diagnostic decision-support capability. This digital simulation layer does not replace physical laboratory assays; rather, it performs Optimal Experimental Design (OED) to prioritize the most informative laboratory tests:

1. Multiphysics PDE Modeling: • 1D Reaction-Diffusion PDE Solver: Solves transient mass transport across the split cotyledon half-thickness (L_half = 1.1 mm) on PyTorch CUDA tensors, coupling aqueous ionic diffusion (Na+, Ca2+, H+), Arrhenius temperature dependence, and stoichiometric middle-lamella pectin de-crosslinking kinetics.

2. Kinetic Identifiability: • Peleg & Fickian Parameter Identifiability: Computes mathematical bounds and parameter sensitivity on water uptake kinetics, establishing whether a 30-minute soak is limited by external surface resistance or internal cellular diffusion.

3. Monte Carlo Uncertainty: • GPU-Accelerated Monte Carlo Sweeps: Executes large-scale parameter sweeps (N = 5,000 to 250,000 scenarios) across uncertain inputs (feed moisture, solids loss, drying temperatures) on an NVIDIA GeForce RTX 4050 Laptop GPU, mapping variance envelopes for thermal duty and dryer sizing.

4. Process Thermodynamic Modeling: • Aspen Plus Electrolyte Property Simulation: Simulates aqueous electrolyte equilibrium (ELECNRTL property method) to predict pH, species activity, and vapor-liquid equilibrium for industrial steep liquor and zero-liquid-discharge evaporator utilities.

5. Dream-RSI Decision Logic: • Dream-RSI Policy Exploration Tree: Implements an autonomous 8-stage discovery and falsification tree (arXiv:2609.14858 / kudoej/dream-rsi) over recorded experimental traces to evaluate multi-objective Pareto trade-offs (cooking time vs recovery vs sodium retention vs cost) without synthetic lookalikes.

> **[MANDATORY EPISTEMIC TAG & COMPUTATIONAL GOVERNANCE]**
> *All computational outputs are formally classified as COMPUTATIONAL_SIMULATION — UNVALIDATED_PRIOR_TO_LAB_DATA. Simulation models serve strictly as exploratory decision support to rank hypotheses and minimize laboratory expenditure. No simulation result will be represented as physical experimental evidence until validated against the client's actual incoming lot.*

6. Technical Investigation Matrix

The multi-module experimental investigation matrix integrates all physical, chemical, structural, and cooking evaluations across the comparative samples:

Module	Analytical Domain	Key Parameters Measured	Standard Method / Instrument	Decision Relevance
MOD-01	Physical & Integrity	Dimensions, 1000-grain weight, bulk density, broken grains, seed-coat %.	IS 4333 / Caliper & Density Flask	Identifies geometric barriers & milling quality.
MOD-02	Proximate Composition	Moisture (WB), crude protein, crude fat, crude fibre, total ash, carbohydrates.	AOAC 925.10 / 920.87 / 922.06 / 923.03	Establishes baseline chemical composition & dry matter.
MOD-03	Starch Properties	Total starch content, amylose percentage, amylopectin ratio.	Megazyme K-AMYL / Spectrophotometry	Assesses starch retrogradation & swelling potential.
MOD-04	DSC Thermal Behavior	Gelatinization onset (To), peak (Tp), conclusion (Tc), enthalpy (ΔHgel).	AACC 56-11.02 / Differential Scanning Calorimeter	Identifies if starch requires higher cooking temperature.
MOD-05	Hydration Kinetics	Water absorption rate (k), equilibrium capacity, leached solids at 25/40/60°C.	AACC 56-20 / Gravimetric Kinetic Cell	Isolates water penetration rate from cooking reactions.
MOD-06	Pectin & Cell-Wall	Water-, oxalate-, alkali-soluble pectin, degree of esterification (DE).	Carbohydrate Polymers / Chemical Fractionation	Tests middle lamella pectin solubility & crosslinking.

Module	Analytical Domain	Key Parameters Measured	Standard Method / Instrument	Decision Relevance
MOD-07	Minerals & Phytate	Calcium, magnesium, iron, phosphorus, potassium, phytic acid, Ca:phytate ratio.	AOAC 999.11 / ICP-OES & Wade Reagent	Quantifies divalent cation availability for pectate bridges.
MOD-08	Hardness & Texture	Raw grain puncture resistance, soaked firmness, peak compression force.	TA.XT Plus Texture Analyzer / 50kg Load Cell	Establishes objective mechanical firmness baseline.
MOD-09	Standardized Cooking	Cooking curve (0–60 min), objective time-to-softness, hard-core rate, solids loss.	Pressurized Vessel / Instrumented Load Cell	Replaces domestic whistle counting with objective kinetics.

6.1 Comprehensive 10-Hypothesis Discrimination Register (Client Feedback A)

To correct the scientific deficiencies of prior drafts (Client Feedback A), pectin–mineral interactions are no longer assumed to explain the cooking behaviour of the client's particular lot. Phase 1 investigates a comprehensive register of ten competing hypotheses. Each hypothesis defines observable predictions, discriminating tests, and experimental boundaries:

Hypothesis	Candidate Mechanism	Observable Predictions & Test	Discriminating Status & Limits
H1	Hydration Kinetics / Diffusion Barrier	Slower water uptake curve, lower initial hydration rate (k), delayed moisture penetration into cotyledon core.	WP-03 gravimetric time-course; separates diffusion limits from cooking limits.
H2	Seed-Coat & Microstructural Resistance	Residual seed-coat fragments, thicker hylar boundary, or dense outer cotyledon cell packing restricting fluid flow.	WP-01 seed-coat ratio & WP-04 light microscopy; evaluates physical milling scouring.
H3	Cell-Wall Pectin & Mineral Crosslinking	Higher insoluble pectin fraction, elevated calcium/magnesium crosslinks, lower phytate-to-mineral molar ratio.	WP-04 pectin fractions & ICP-OES minerals; plausible biochemical hypothesis.
H4	Starch Gelatinization & Thermal Impedance	Elevated DSC gelatinization peak temperature (Tp > 78°C) or higher transition enthalpy requiring prolonged heat input.	WP-02 DSC thermal assay; identifies intrinsic starch cooking resistance.
H5	Cultivar Genetics & Agronomic Origin	Distinct genetic variety, protein matrix density, or soil mineral accumulation unique to East African agro-climates.	Requires sample lot metadata; Phase 1 identifies phenotypic expression only.
H6	Commercial Dehulling & Processing History	Thermal case-hardening from dry milling, excessive frictional heat during emery scouring, or residual polishing oil.	Surface lipid assay & microstructural examination; identifies commercial processing damage.
H7	Storage Age & Hard-to-Cook (HTC) Defect	Prolonged transit/storage under high temperature (>30°C) and RH (>70%) inducing phytase loss and pectin crosslinking.	Standard legume aging marker; correlated with storage records and ion leakage.

Hypothesis	Candidate Mechanism	Observable Predictions & Test	Discriminating Status & Limits
H8	Grain Physical Geometry & Split Uniformity	Larger cotyledon thickness, lower surface-area-to-volume ratio, or higher broken grain fraction skewing water distribution.	WP-01 caliper dimensions and 1000-grain mass; isolates geometric confounders.
H9	Cooking Apparatus & Measurement Confounders	Apparatus heat transfer, pressure vessel venting variability, or whistle-counting bias in domestic observation.	WP-05 standardized cooking protocol eliminates domestic apparatus artefacts.
H10	Multi-Factorial Synergistic Interaction	Simultaneous interaction of slow hydration (H1) and cell-wall pectin resistance (H3) amplifying total cooking latency.	Multivariate correlation & regression modeling in WP-05 final synthesis.

Evidence Language Ladder: To maintain scientific integrity, Phase 1 conclusions strictly observe the following four-tier hierarchy: (1) Measured Difference: Directly observed analytical variance between samples; (2) Statistical Association: Correlation supported by data but not establishing directionality; (3) Plausible Hypothesis: Literature-compatible mechanism not yet causally proven for the target lot; (4) Controlled Causal Effect: Established only when controlled intervention trials isolate the mechanism in Phase 2.

7. Food Safety, Regulatory Separation & Product Acceptance Criteria

Under project governance standards, statutory food safety regulations are formally segregated from internal commercial product quality specifications. The commercial product falls under FSSAI Category 2.4.6.15 (Dehulled Split Pulses):

Quality / Safety Parameter	Classification & Authority	Numerical Standard / Limit	Measurement Method & Stage
Moisture Content	FSSAI Statutory Standard (Clause 2.4.6.15)	Maximum 14.0% w/w (Wet Basis)	AOAC 925.10 / Mandatory release check
Broken Grains / Fines	FSSAI Statutory Standard (Clause 2.4.6.15)	Maximum 5.0% w/w broken dal	IS 4333 Part 1 / Sieve analysis
Project Broken Target	Project Acceptance Target (Internal Quality)	Maximum 2.0% w/w broken dal	Stricter internal commercial specification
Finished Sodium Content	Project Acceptance Target (Commercial Quality)	Maximum 150 mg / 100 g dry dal	ICP-OES / Evaluated post-drying in Phase 2
Water Activity (aw)	Project Acceptance Target (Microbial Stability)	aw < 0.60 at 10.0% ± 0.5% moisture	Chilled mirror dew-point hygrometer

Laboratory Safety Rule: No human sensory evaluation or tasting of uncharacterized experimental samples will occur during Phase 1. Formal organoleptic panels are restricted to Phase 2 under certified food-grade hygiene protocols.

8. Key Deliverables & Final Technical Report Structure

FinnoaQ will provide the complete suite of 14 key deliverables defined in Section 21 of the source SOW:

Deliverables 1 to 7 (Analytical Modules)	Deliverables 8 to 14 (Synthesis & Reports)

Deliverables 1 to 7 (Analytical Modules)	Deliverables 8 to 14 (Synthesis & Reports)
DEL-01: Sample Characterization Report	DEL-08: Hardness & Texture Analysis Report
DEL-02: Proximate Chemical Composition Report	DEL-09: Standardized Cooking Performance Report
DEL-03: Starch & Gelatinization Report	DEL-10: Tanzanian vs. Indian Benchmarking Matrix
DEL-04: Hydration & Water Absorption Report	DEL-11: Integrated Root-Cause Analysis Dossier
DEL-05: Structural & Seed-Coat Assessment	DEL-12: Phase 2 Technology Recommendations
DEL-06: Pectin & Cell-Wall Fractionation Report	DEL-13: Executive Management Summary Briefing
DEL-07: Mineral & Phytate Assessment Report	DEL-14: Comprehensive Final Technical R&D Report

Final Technical Report Structure: The final deliverable comprises a comprehensive 23-section technical dossier covering executive summary, background, experimental design, analytical methodologies, detailed results for all modules, benchmark matrices, statistical analyses, root-cause assessment, Phase 2 roadmap, study limitations, and raw data appendices.

9. Project Timeline & Operational Milestones

The indicative turnaround is 20–25 business days across five operational stages (source SOW Section 27):

Stage	Timeline	Key Operational Activities	Dependencies & Deliverables
Stage 1	Days 1–3	Sample intake, physical inspection, chain of custody coding, allocation to internal/external labs.	Sample receipt + Milestone 1 advance
Stage 2	Days 4–14	Proximate chemical analysis, starch/amylose assays, DSC thermal runs, mineral/phytate testing.	Completion of Stage 1 preparation
Stage 3	Days 10–18	Time-course hydration kinetics, standardized pressurized cooking runs, texture analyzer shear testing.	Conducted in parallel with Stage 2
Stage 4	Days 18–22	Data consolidation, statistical evaluation, cross-verification, multivariate correlation, root-cause synthesis.	Receipt of all analytical data sheets
Stage 5	Days 22–25	Final technical report compilation, executive briefing preparation, submission, and client presentation.	Final report sign-off + Milestone 3

10. Sample Requirements, Allocation Budget & Multi-Lot Options

Under Section 3 of the source SOW, the recommended sample mass is approximately 2.5–3.0 kg per lot. A rigorous physical sample allocation budget was performed across all analytical modules:

Module / Allocation	Analytical Activity & Replicate Structure	Mass Budget (g)	Allocation Basis
MOD-01	Physical & Preliminary Grain Integrity Characterization	150 g	3 batches x 3 repeats + reserve
MOD-02	Proximate Chemical Composition (Moisture, Protein, Fat, Ash, Fibre)	250 g	Triplicate AOAC external lab minimum
MOD-03	Starch & Amylose / Amylopectin Spectrophotometric Assay	150 g	Triplicate Megazyme assay + reserve
MOD-04	DSC Thermal Gelatinization & Transition Enthalpy	150 g	Triplicate micro-sample DSC runs
MOD-05	Hydration Kinetics & Time-Course Water Absorption (25°C, 40°C, 60°C)	300 g	Full kinetic curves across 3 temperatures
MOD-06	Pectin & Cell-Wall Chemical Fractionation	150 g	Triplicate chemical sequential extraction
MOD-07	Mineral & Phytate Profile (ICP-OES / AAS)	150 g	External accredited laboratory minimum
MOD-08	Grain Hardness & Structural Texture Analysis (Load Cell)	150 g	Raw & soaked texture profile runs
MOD-09	Standardized Pressurized Cooking Performance Study	400 g	Multiple pressurized cooking runs
SEC-RET	Sealed Archival Reference Retention Sample (Dispute & Verification)	500 g	Mandatory ISO 17025 climate archive
SEC-CON	Grinding Loss, Handling Contingency & Repeat Margin (~15%)	300 g	Milling hold-up and operational buffer
TOTAL LOT BUDGET	**Total Requisite Sample Mass per Grain Lot**	**2,550 g (2.55 kg)**	**Strictly fits within 2.5–3.0 kg allocation**

Multi-Lot Commercial Options: To provide complete commercial transparency, FinnoaQ offers two structured options for client approval:

Option 1: • Option 1 (Baseline Scoped SOW — ₹1,73,000 + GST): Two traceable commercial samples (1 Tanzanian Lot + 1 Indian Benchmark Lot), 2.5–3.0 kg each. Provides rigorous, lot-specific diagnostic comparison within the original budget and 20–25 business day timeline.

Option 2: • Option 2 (Multi-Lot Expanded Variation — ₹2,85,000 + GST): Four independently traceable commercial lots (2 Tanzanian import consignments + 2 Indian regional benchmarks), 2.5–3.0 kg each. Requires doubling analytical laboratory runs, expanding inter-lot statistical variance modeling, and an extended 30–35 business day timeline.

11. Experimental Design, Replication Hierarchy & Statistical Plan

Experimental units adhere strictly to the four-tier statistical governance hierarchy:

Tier 1: • Tier 1: Independent Commercial Grain Lot (distinct physical consignment with verifiable provenance).

Tier 2: • Tier 2: Independent Preparation Batch (weighed, hydrated, or processed in separate vessels on separate occasions, n = 3).

Tier 3: • Tier 3: Technical Replicate (repeated instrumental measurement of the same prepared batch, n = 3).

Tier 4: • Tier 4: Subsample (analytical split of a single batch; never conflated with independent lots).

Statistical Testing Discipline: All analytical data are reported as Mean ± Standard Deviation with 95% Confidence Intervals. Differences between Tanzanian dal and Indian benchmark are evaluated using two-sample Welch's t-test (accounting for unequal variances) and one-way ANOVA with Tukey's HSD post-hoc test.

12. Future Treatment-Candidate Pathway (Phase 2 Gated Scope)

All chemical conditioning agents and physical treatments are rigorously segregated. Reagents are identified by precise chemical formula, CAS registry, food additive INS number, and concentration basis:

Treatment ID	Reagent & Formula	Concentration & Dose	Process Parameters	Regulatory & Phase Status
TRT-00	Untreated Control (Raw As-Received)	0.00% chemical 0.00% grain dose	Ambient (25°C), 0 min; as-received 10.5% WB.	FSSAI 2.4.6.15 compliant; PHASE 1 DIAGNOSTIC BASELINE
TRT-01	Water-Only Soak Control (Pure H_2O, CAS 7732-18-5)	0.00% chemical Liquor ratio 3:1	25°C, 60 min, RO rinse 1:1; tray dry 55°C to 10% WB.	Pure potable water; PHASE 2 SCREENING CONTROL
TRT-02	Sodium Bicarbonate ($NaHCO_3$, CAS 144-55-8)	0.25%–0.50% w/w bath (0.75%–1.50% as-fed grain)	25°C, 60 min, pH 8.2–8.4; rinse 2:1, dry to 10% WB.	INS 500(ii) food additive; PHASE 2 CANDIDATE — NOT TESTED
TRT-03	Trisodium Citrate ($Na_3C_6H_5O_7 \cdot 2H_2O$, CAS 6132-04-3)	0.50% w/w bath (1.50% w/w as-fed grain)	25°C, 60 min, pH 7.9–8.2; rinse 2:1, dry to 10% WB.	INS 331(iii) sequestering agent; PHASE 2 CANDIDATE — NOT TESTED
TRT-04	Citrate Buffer System (Citrate + Citric Acid)	0.40% citrate + 0.05% acid (1.20% + 0.15% as-fed grain)	25°C, 60 min, buffer pH 6.0; rinse 2:1, dry to 10% WB.	INS 331(iii) + INS 330; PHASE 2 CANDIDATE — NOT TESTED
TRT-05	Sodium Carbonate (Segregated) (Na_2CO_3, CAS 497-19-8)	0.20% w/w bath (0.60% w/w as-fed grain)	25°C, 45 min, pH 10.6–10.9; rinse 2.5:1, dry to 10% WB.	INS 500(i) alkaline salt; PHASE 2 CANDIDATE — NOT TESTED
TRT-06	Hydrothermal Pre-Steeping (Pure Water, Zero Chemical)	0.00% chemical Physical thermal process	65°C, 20 min steeping; fluidized bed dry to 10% WB.	100% clean-label physical process; PHASE 2 CANDIDATE — NOT TESTED

13. Indicative Technical Cost Structure & Commercial Terms

The indicative professional fee totals exactly ₹1,73,000 plus applicable GST, matching Table 40 of the source SOW:

Item	Analytical Module / Professional Service Component	Indicative Fee (INR)
01	Sample Preparation & Physical Characterization (Dimensions, 1000-grain weight, density)	₹20,000
02	Proximate Chemical Analysis (Moisture, protein, fat, fibre, ash, carbohydrates)	₹7,000

Item	Analytical Module / Professional Service Component	Indicative Fee (INR)
03	Starch & Amylose Analysis (Total starch, amylose/amylopectin ratio)	₹5,000
04	DSC Thermal / Gelatinization Characterization (To, Tp, Tc, ΔHgel)	₹20,000
05	Hydration & Water Absorption Study (Kinetics at 25°C, 40°C, 60°C across 0–360 min)	₹10,000
06	Pectin & Cell-Wall Characterization (Water-, oxalate-, alkali-soluble fractions, DE)	₹18,000
07	Mineral & Phytate Profile (Ca, Mg, Fe, P, K by ICP-OES/AAS, phytic acid content)	₹16,000
08	Hardness & Texture Analysis (Raw & soaked load-cell puncture, shear force)	₹17,000
09	Standardized Cooking Performance Study (Pressurized kinetics, hard-core rate)	₹20,000
10	Integrated Data Analysis & Root-Cause Assessment (Multivariate hypothesis testing)	₹20,000
11	Final Technical R&D Report & Executive Management Summary Compilation	₹20,000
TOTAL	**TOTAL PHASE 1 INDICATIVE PROFESSIONAL FEE (EXCLUDING GST)**	**₹1,73,000**

Commercial Notes: (1) Total professional fee is ₹1,73,000 plus applicable GST; (2) Commercial ceiling is up to ₹2,00,000 + GST; no additional charges will be incurred without prior written client approval; (3) Third-party specialized laboratory assays are coordinated within this budget framework; (4) GST is payable extra on actuals under SAC 998313.

Milestone Payment Schedule (Source SOW Table 45)

Milestone	Objective Deliverable Trigger	Percentage	Amount (INR)
Milestone 1	Project Initiation: Formal confirmation & sample receipt protocol	40.0%	₹69,200 + GST
Milestone 2	Mid-Term Analytical Completion: Delivery of laboratory data sheets	40.0%	₹69,200 + GST
Milestone 3	Final Report Acceptance: Delivery of Final Technical Report & Briefing	20.0%	₹34,600 + GST

Milestone	Objective Deliverable Trigger	Percentage	Amount (INR)
TOTAL	**Consolidated Project Milestone Value**	**100.0%**	**₹1,73,000 + GST**

14. Assumptions, Exclusions & Risk Management

Key operational assumptions and risk mitigations governing execution:

Assumption 1: • Sample Custody & Authenticity: Client is responsible for supplying genuine, traceable Tanzanian dal and Indian benchmark dal with completed Sample Metadata Forms. Delays in sample arrival will shift project milestones accordingly.

Assumption 2: • External Laboratory Throughput: Specialized testing at candidate partner institutions is subject to instrument queue times. Any laboratory delays will be communicated immediately without cost escalation.

Risk Mitigation: • Diagnostic Risk Control: If preliminary assays reveal anomalous composition (e.g. extreme moisture deviation or heavy mold infestation), testing will pause immediately for client review before exhausting budget.

15. Confidentiality, Data Ownership & Sample Retention

• Client Data Ownership: Under Section 31 of the source SOW, all experimental data, laboratory test certificates, statistical models, and final reports generated under Phase 1 remain the exclusive intellectual and commercial property of the Client.

• Non-Disclosure Obligations: FinnoaQ treats all sample identities, commercial origins, pricing data, and technical findings as strict commercial-in-confidence material. No publication, presentation, or external dissemination will occur without prior written authorization from the Client.

• Sample Retention: Unused grain material and a 500 g sealed reference retention sample per lot will be retained under controlled climate storage (18–22°C, dry conditions) for six months following report submission for verification or dispute resolution.

16. Project Acceptance & Client Onboarding Signature Block

To initiate the Phase 1 Comparative Diagnostic R&D Study, please endorse the authorization block below and return a signed copy to FinnoaQ Agro-Industrial Technologies Ltd.

CLIENT ACCEPTANCE & AUTHORIZATION	EXECUTING R&D CONTRACTOR CONFIRMATION
Client Entity: Technical Advisory Panel	Contractor: FinnoaQ Agro-Industrial Technologies Ltd.
Authorized Signatory: Technical Advisory Panel	Authorized Lead Engineer: Sourish Senapati
Designation: CEO & Director	Designation: Lead Chemical & Process Engineer
Scope Selection: [ ] Option 1 (₹1.73L) [ ] Option 2 (₹2.85L)	Technical Guidance: FinnoaQ Process Engineering Board
Signature & Date: _____	Signature & Date: _____

Annexure A: Laboratory Selection Criteria & Candidate Partner Institutions

Under Section 23 of the source SOW, specialized analytical modules requiring accredited analytical equipment will be coordinated with candidate institutions based on technical capability, method validity, turnaround time, and competitive quotation. These institutions are candidates for quotation and evaluation—not pre-committed subcontractors:

1. • Eureka Analytical Services Pvt. Ltd. (Kolkata / Bangalore): NABL accredited food testing facility; candidate for proximate composition, heavy metals, pesticide residues, and mineral profiling via ICP-OES.

2. • Mitra S.K. Private Limited (Kolkata): NABL / FSSAI notified laboratory with extensive commodity testing infrastructure; candidate for proximate chemical analysis, fiber fractionation, and grain physical parameters.

3. • CSIR-Central Glass & Ceramic Research Institute (CGCRI, Kolkata): Premier central research institute; candidate for advanced materials characterization, scanning electron microscopy (SEM), and thermal analysis.

4. • Department of Chemical Engineering, IIT Kharagpur: Premier academic institution; candidate for advanced biophysical characterization, differential scanning calorimetry (DSC), and texture profile analysis under research collaboration protocols.

Annexure B: Provisional 15,000 kg/day Plant Concept & Independent Mass-Balance Audit (Client Feedback B)

To resolve Client Feedback B, this annexure provides the complete mathematical and thermodynamic derivation of the provisional 15,000 kg/day industrial processing concept. The plant concept is strictly a downstream conceptual scenario; final engineering sizing is gated upon Phase 1 and Phase 2 empirical data.

B.1 Mathematical Derivation & Basis Reconciliation

Incoming Raw Grain: Total feed M_feed = 15,000.000 kg/day at 10.5% wet-basis moisture (X_wb = 0.105). Dry solids M_ds,in = 15,000 x (1 - 0.105) = 13,425.000 kg/day. Intrinsic grain water M_water,in = 1,575.000 kg/day.

Solids Loss Audit: One reported assumption states an absolute solids loss of 184.000 kg/day, leaving 13,241.000 kg/day dry solids. This loss represents exactly 1.370577% (approx. 1.3706%) of incoming dry solids. An alternate assumption in earlier notes stated a 1.46% solids loss, which would equal 13,425 x 0.0146 = 196.005 kg/day loss, leaving 13,228.995 kg/day dry solids. These two assumptions are mathematically distinct. For this reconciliation, the audited documented solids value of 13,241.000 kg/day is adopted.

Final Dried Product: Commercial product target moisture is 10.0% wet-basis (X_wb = 0.100). Total product mass M_prod = 13,241.000 / (1 - 0.100) = 14,712.222 kg/day, containing 1,471.222 kg/day water.

B.2 Reconciled Pre-Dryer & Evaporation Scenarios

Because post-drainage surface liquid retention has not yet been measured for the target Tanzanian lot, two competing scenarios are reconciled:

Mass & Moisture Parameter	Scenario B1 (True 28.0% WB)	Scenario B2 (Reported 18,590 kg/day)
Pre-Dryer Total Wet Stream Mass	18,390.278 kg/day	18,590.000 kg/day
Pre-Dryer Dry Solids Mass	13,241.000 kg/day	13,241.000 kg/day
Pre-Dryer Water Content	5,149.278 kg/day	5,349.000 kg/day
Pre-Dryer Wet-Basis Moisture Content	28.0000% w/w	28.7735% w/w (Implied)
Final Product Water (10.0% WB)	1,471.222 kg/day	1,471.222 kg/day
Dryer Water Evaporation Required	3,678.056 kg/day (183.90 kg/h)	3,877.778 kg/day (193.89 kg/h)

Thermodynamic Duty Impact: Scenario B2 requires evaporating 199.722 kg/day more water (+5.43%) than Scenario B1. At an effective heat of vaporization of 2,400 kJ/kg, continuous thermal drying duty increases from 122.60 kW_th (Scenario B1) to 129.26 kW_th (Scenario B2), increasing steam demand by 12.9 kg/h. Status of Finding B is UNRESOLVED — SCENARIOS SHOWN; final dryer sizing is held provisional until bench-scale hydration and drainage are measured.

Annexure C: Sample Custody & Metadata Registration Form

Please complete and enclose this verification form with the dispatched sample consignments:

Sample Unique Identifier:	[To be assigned upon receipt at FinnoaQ Lab]
Commercial Origin & Cultivar:	[e.g. Tanzanian Arusha / Indian Latur Desi]
Consignment / Lot Number:	[Bill of Lading / Invoice Batch Number]
Date of Dehulling / Milling:	[DD/MM/YYYY or estimated month]
Import Date & Port of Entry:	[Applicable for imported consignments]
Polishing / Surface Treatment:	[e.g. Water polish / Oil polish / Unpolished]
Storage History & Packaging:	[e.g. 50 kg PP woven bags / Climate controlled]
Authorized Dispatch Contact:	[Name, Designation, Phone, Email]

Annexure D: Scientific References & Regulatory Citations

1. AOAC International. (2019). Official Methods of Analysis of AOAC International (21st ed.). Gaithersburg, MD: AOAC International.

2. Bureau of Indian Standards. (2014). IS 4333: Methods of Analysis for Foodgrains. New Delhi: BIS.

3. Food Safety and Standards Authority of India (FSSAI). (2011). Food Safety and Standards (Food Products Standards and Food Additives) Regulations. Compendium of Food Additives & Standard 2.4.6.15 (Pulses).

4. Hincks, M. J., & Stanley, D. W. (1986). Multiple mechanisms for the hard-to-cook defect in dry beans. Journal of Food Science, 51(3), 738-742.

5. Jones, P. M. B., & Boulter, D. (1983). The cause of reduced cooking quality in stored cowpeas. Journal of Food Science, 48(2), 629-634.

6. Sefa-Dedeh, S., Stanley, D. W., & Voisey, P. W. (1979). Effect of storage conditions on cooking quality and microstructure of cowpeas. Journal of Food Science, 44(5), 1309-1314.

7. Tiwari, B. K., Gowen, A., & McKenna, B. (2011). Pulse Foods: Processing, Quality and Nutritive Value. Academic Press.

Annexure E: Source SOW 36-Section Traceability & Concordance Matrix

Concordance audit verifying that all 36 numbered sections from the source SOW (Phase1_Scope_of_Work_Toor_Dal.docx) are substantively preserved:

Source Sections	Source Scope Domain	Revised Proposal Location & Status
Sections 1–3	Background, Objectives, Samples	Sections 1, 2, 3, 10 (Preserved 100%; whistle context qualified)
Sections 4–7	Approach, Physical, Proximate, Starch	Sections 5, 6 (WP-01, WP-02, WP-03, simulation methodology incorporated)
Sections 8–11	Hydration, Structure, Pectin, Minerals	Section 6 (WP-04, WP-05, WP-06, WP-07 incorporated)
Sections 12–15	Texture, Standardized Cooking Design	Sections 6, 11 (WP-08, WP-09, instrumented endpoint defined)

Source Sections	Source Scope Domain	Revised Proposal Location & Status
Sections 16–18	Root-Cause Pathway & Phase 2 Tech	Sections 6.1, 12 (10-hypothesis register, distinct reagents)
Sections 19–20	QC, Statistics, Integrated Analysis	Sections 6, 11 (Replication hierarchy, Welch's t-test, ANOVA)
Sections 21–23	Deliverables, Report Outline, Labs	Section 8, Annexure A (All 14 deliverables mapped)
Sections 24–29	Fees, Ceiling, Timeline, Payments	Sections 9, 13 (₹1,73,000 line items, ₹2L ceiling, 40/40/20%)
Sections 30–36	Retention, NDA, Limits, Contacts	Sections 4, 14, 15, 16 (Client data ownership, client authorization block)

Concordance Summary: 36 of 36 sections mapped with zero substantive omission. All scope boundaries strictly protected.